

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459404.7166 +/- 0.0071 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.169 +/- 0.021
Transit depth (Rp/Rs)^2	2.87 +/- 0.7 [%]
Orbital Inclination (inc)	84.48 +/- 1.87
Airmass coefficient 1 (a1)	1.016 +/- 0.023
Airmass coefficient 2 (a2)	-0.02 +/- 0.063
Scatter in the residuals of the lightcurve fit is	2.29 %
Best Comparison Star	#3 - [204, 310]
Optimal Aperture	2.91
Optimal Annulus	7.81
Transit Duration (day)	0.066 +/- 0.031

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.169 ± 0.021 vs 0.1394 (deviation 1.35σ)
Duration fit vs published	0.066 d vs 0.0983 d (deviation 1.0σ)
Mid-time O–C	19.2 min
Depth SNR	7.7
Residual scatter	2.29 %

Light curve

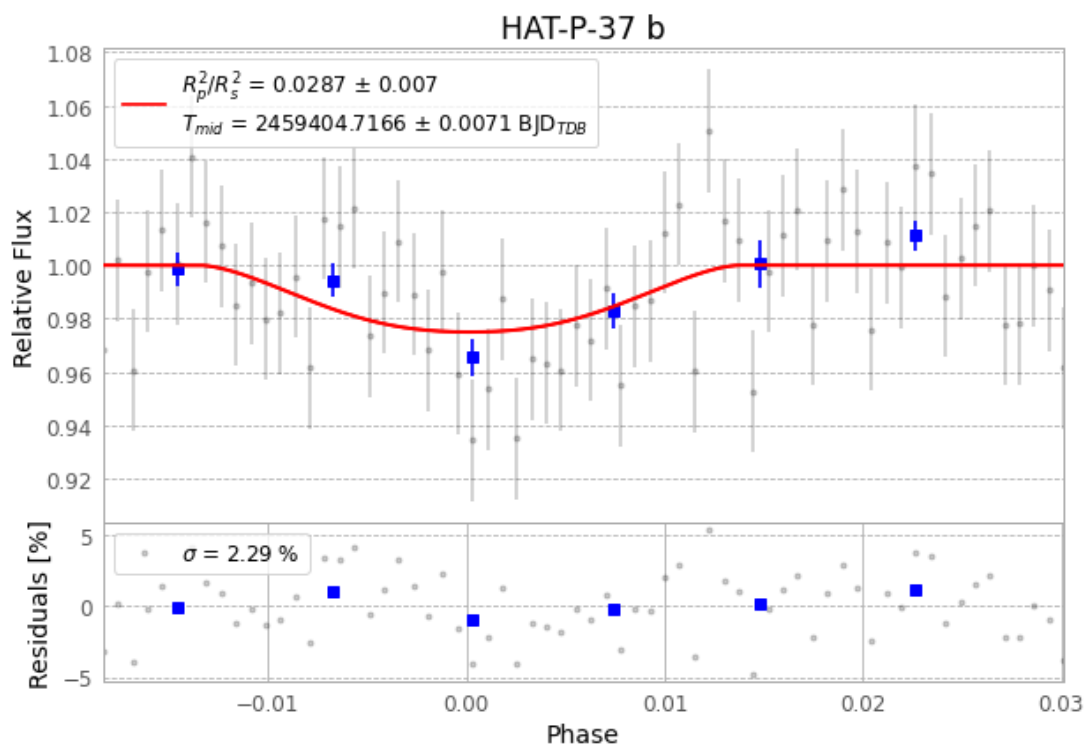


Figure 1 — FinalLightCurve_HAT-P-37 b_2021-07-08.png (SHA-256 d6d734c496aded5b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [336, 253], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 e889dbfe639f0700...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

66 FITS frames (43 MB), HATP-37210709035412.FITS → HATP-37210709070912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-37-210709.log (700ca84510f1...), FinalLightCurve_HAT-P-37 b_2021-07-08.png (d6d734c496ad...), FinalLightCurve_HAT-P-37 b_2021-07-08.csv (030403360bf9...)

Compute resources

Wall time 102.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-27 00:21Z.